

Sensor Array

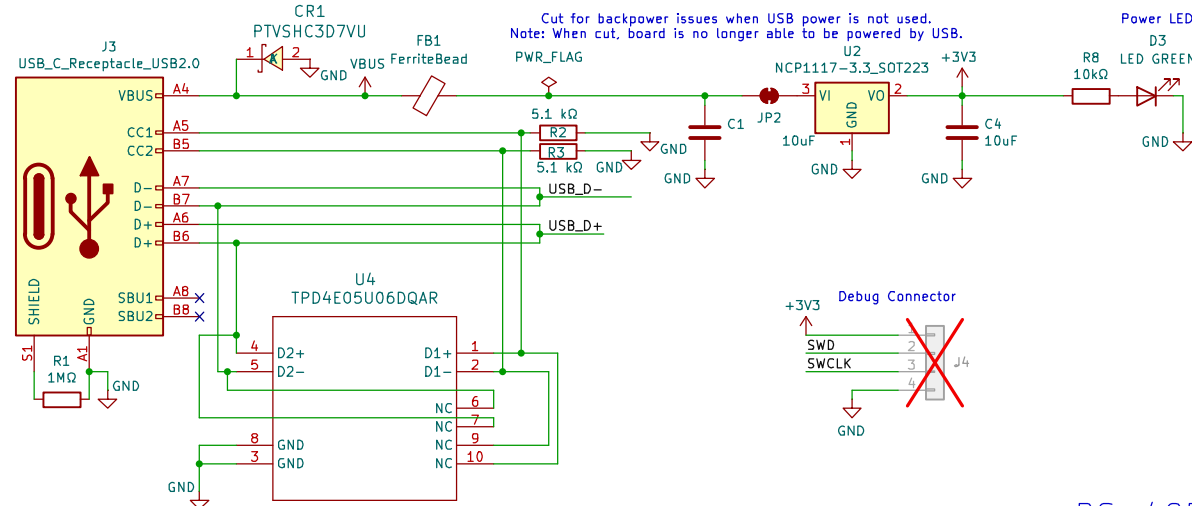
Sensor Array

SPI0_MISO	SPI0_MISO
SPI0_MOSI	SPI0_MOSI
SPI0_SCLK	SPI0_SCLK
SPI1_MISO	SPI1_MISO
SPI1_MOSI	SPI1_MOSI
SPI1_SCLK	SPI1_SCLK
BANK0_CS	BANK_0_CS
BANK1_CS	BANK_1_CS
BANK2_CS	BANK_2_CS
BANK3_CS	BANK_3_CS
BANK4_CS	BANK_4_CS
BANK5_CS	BANK_5_CS
BANK6_CS	BANK_6_CS
BANK7_CS	BANK_7_CS

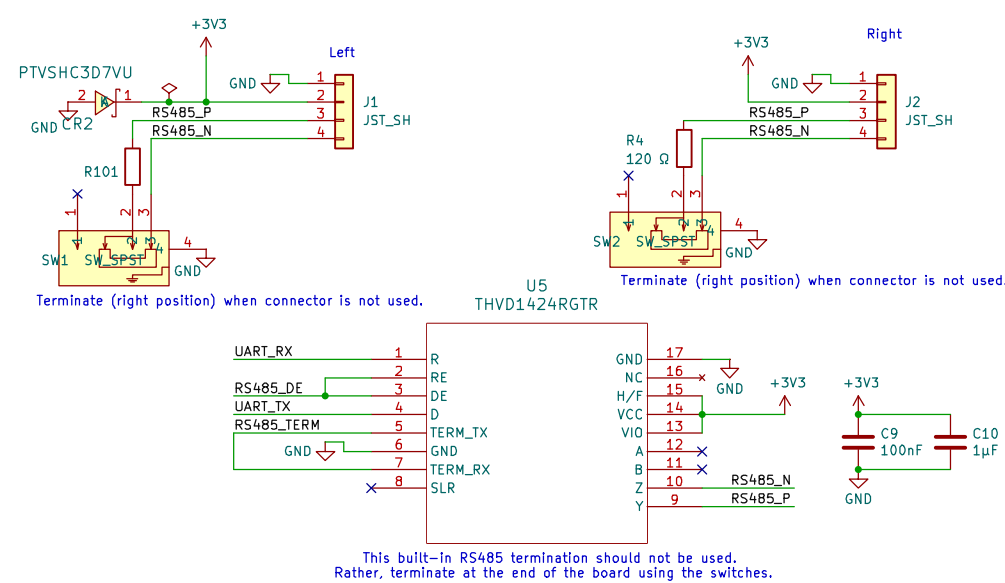
File: Sensor_Array.kicad_sch

- H1 MountingHole
- H2 MountingHole
- H3 MountingHole

Power and USB

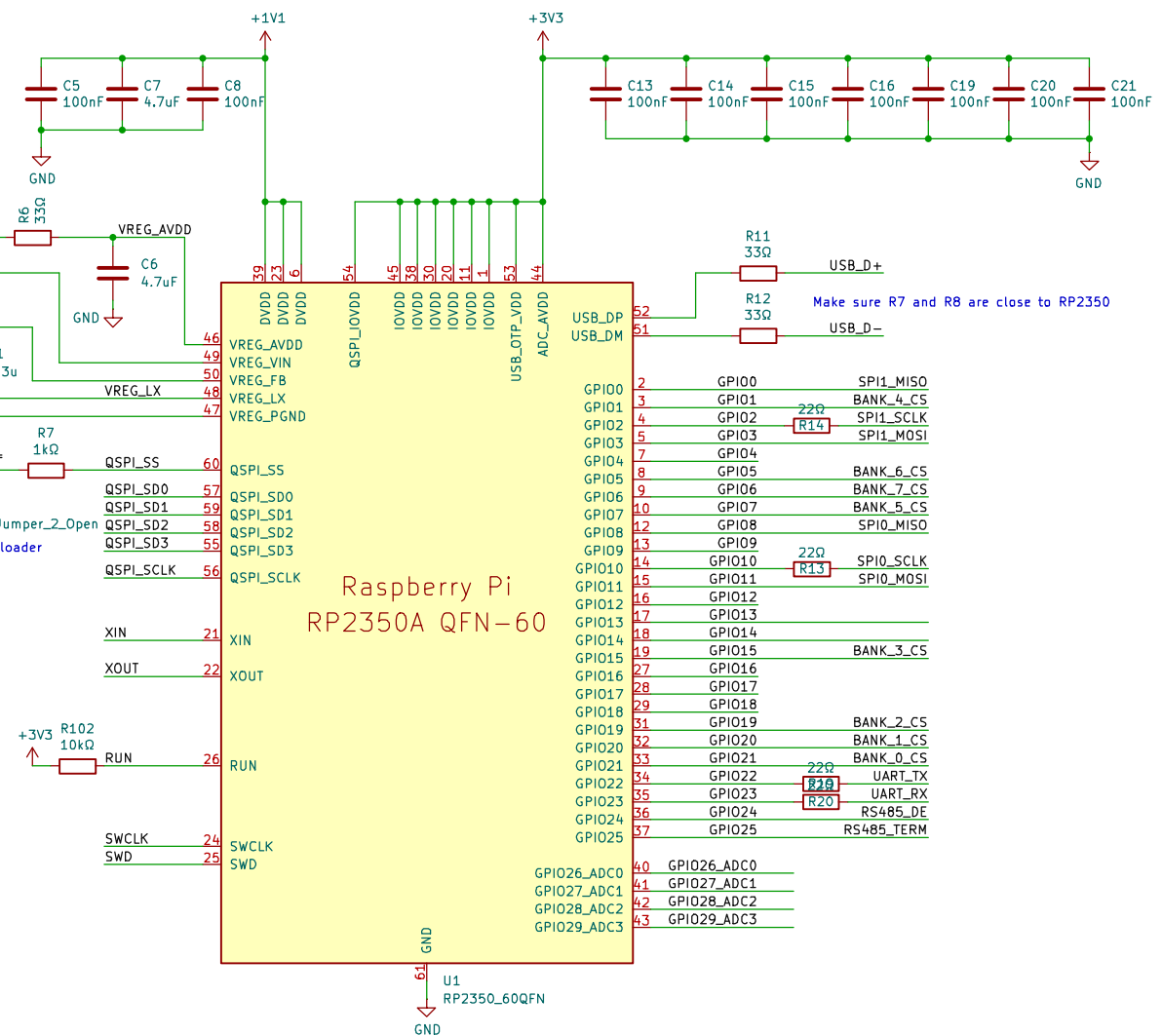


RS-485 Interface

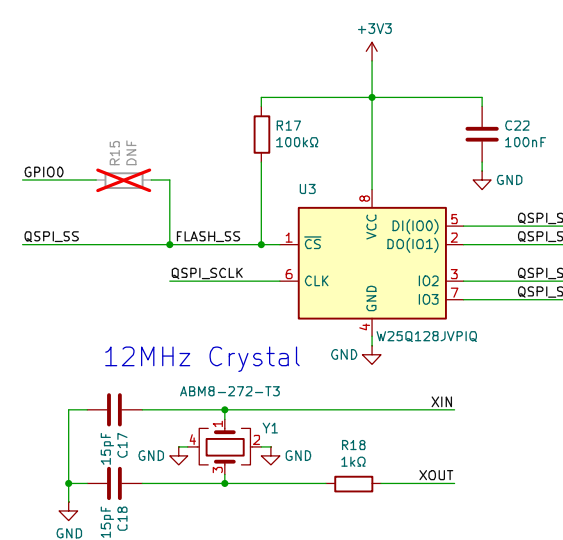


This built-in RS485 termination should not be used. Rather, terminate at the end of the board using the switches.

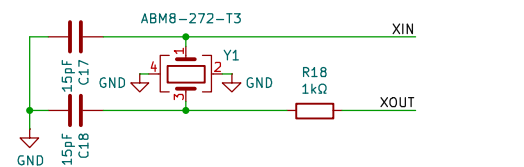
Raspberry Pi RP2350A QFN-60



Flash



12MHz Crystal



The (up-to) 32-sensor array consists of banks of 4.

Bank 0: Sensors 0–3

4_Sensor_bank

SPI_SCLK < SPI0_SCLK
SPI_MOSI < SPI0_MOSI
SPI_MISO >> SPI0_MISO
BANK_CS < BANK0_CS

File: 4_Sensor_Bank.kicad_sch

4_Sensor_bank4

SPI_SCLK < SPI1_SCLK
SPI_MOSI < SPI1_MOSI
SPI_MISO >> SPI1_MISO
BANK_CS < BANK4_CS

File: 4_Sensor_Bank.kicad_sch

4_Sensor_bank1

SPI_SCLK < SPI0_SCLK
SPI_MOSI < SPI0_MOSI
SPI_MISO >> SPI0_MISO
BANK_CS < BANK1_CS

File: 4_Sensor_Bank.kicad_sch

4_Sensor_bank5

SPI_SCLK < SPI1_SCLK
SPI_MOSI < SPI1_MOSI
SPI_MISO >> SPI1_MISO
BANK_CS < BANK5_CS

File: 4_Sensor_Bank.kicad_sch

4_Sensor_bank2

SPI_SCLK < SPI0_SCLK
SPI_MOSI < SPI0_MOSI
SPI_MISO >> SPI0_MISO
BANK_CS < BANK2_CS

File: 4_Sensor_Bank.kicad_sch

4_Sensor_bank6

SPI_SCLK < SPI1_SCLK
SPI_MOSI < SPI1_MOSI
SPI_MISO >> SPI1_MISO
BANK_CS < BANK6_CS

File: 4_Sensor_Bank.kicad_sch

4_Sensor_bank3

SPI_SCLK < SPI0_SCLK
SPI_MOSI < SPI0_MOSI
SPI_MISO >> SPI0_MISO
BANK_CS < BANK3_CS

File: 4_Sensor_Bank.kicad_sch

Last_sensor_bank

SPI_SCLK < SPI1_SCLK
SPI_MOSI < SPI1_MOSI
SPI_MISO >> SPI1_MISO
BANK_CS < BANK7_CS

File: last_sensor_bank.kicad_sch

Note: This sensor bank does *not* inherit from 4_Sensor_bank
This is to allow for DNP sensors, so the total number of sensors
can be a number not modulus 4

SPI Sensor Bank Connections

Sheet: /Sensor Array/
File: Sensor_Array.kicad_sch

Title: 8x4-Channel LED Sensor Array

Size: A4	Date: 2026-01-02	Rev: C
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SPI Sensor Bank Connections

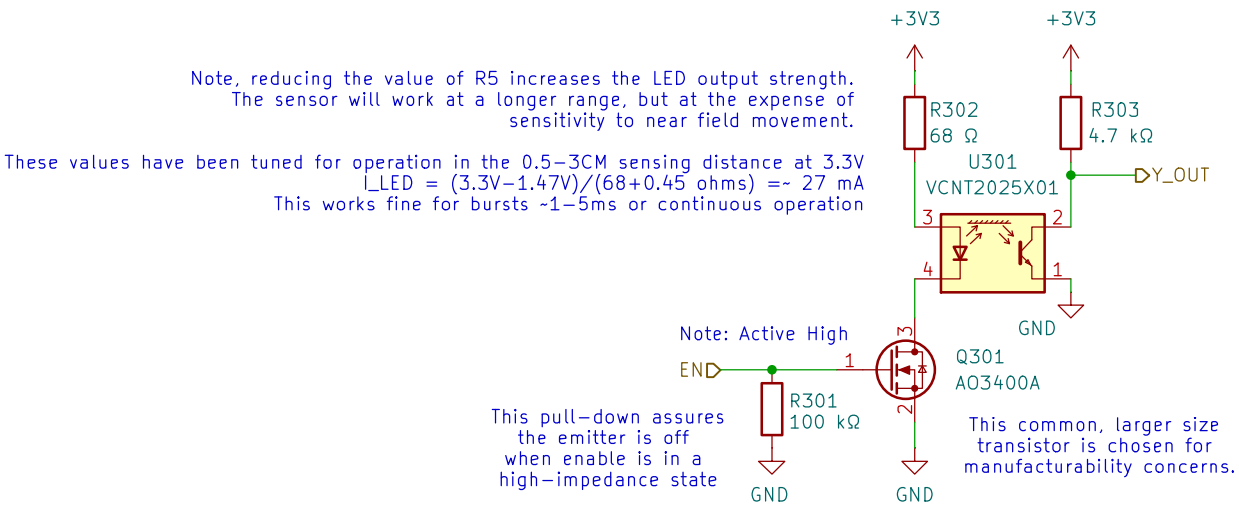
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File: Sensor_Array.kicad_sch

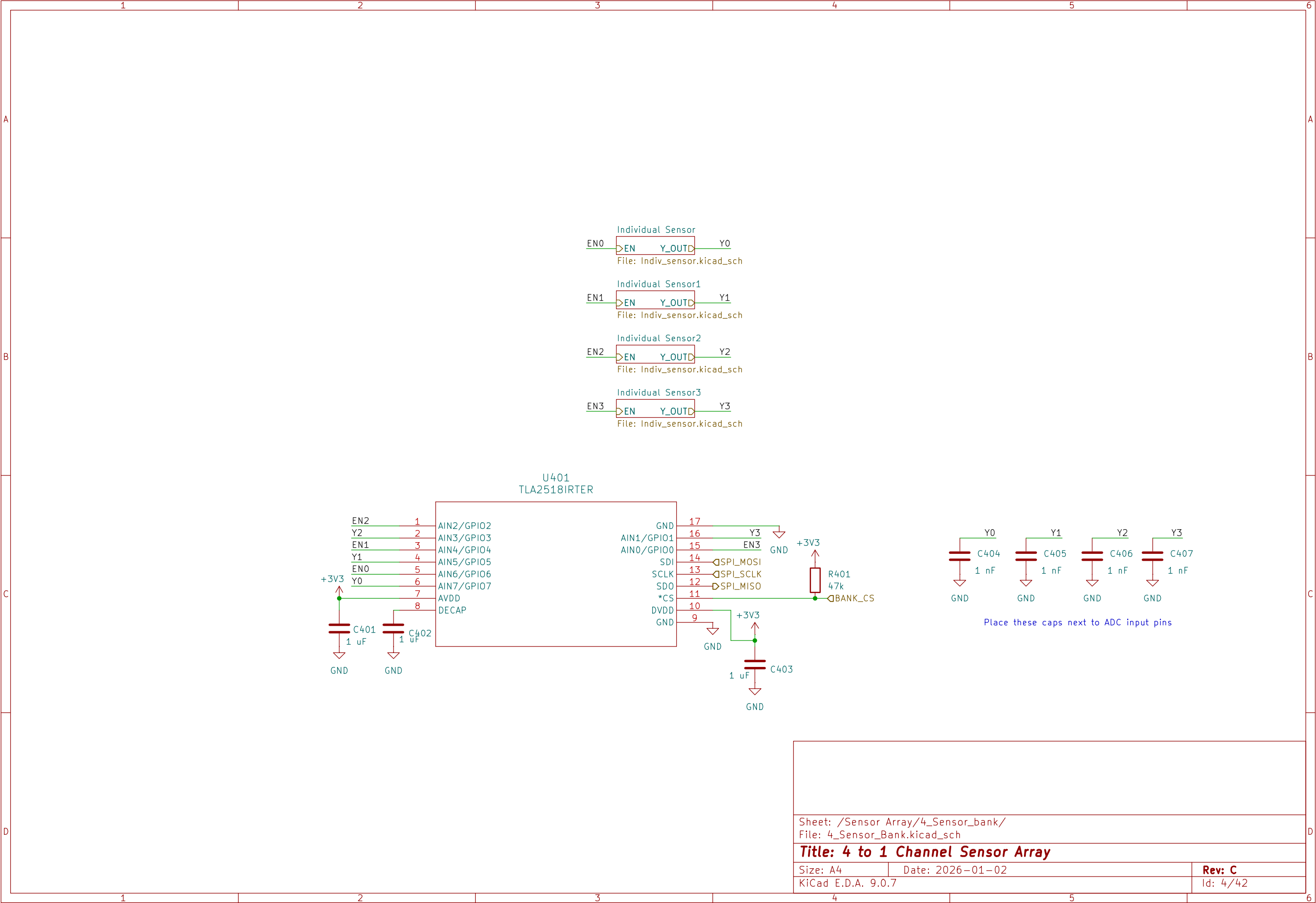
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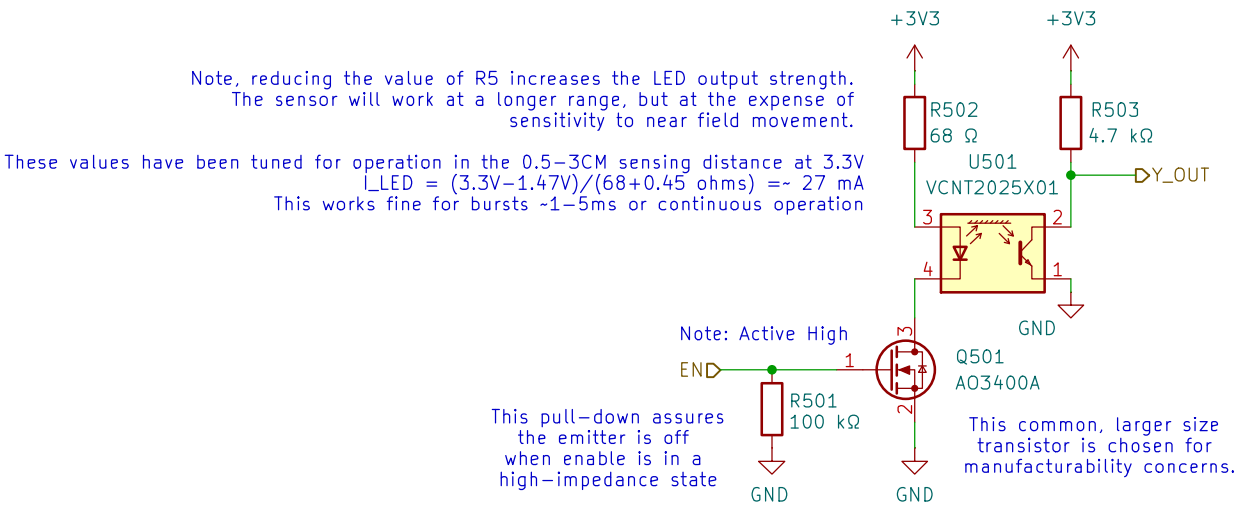
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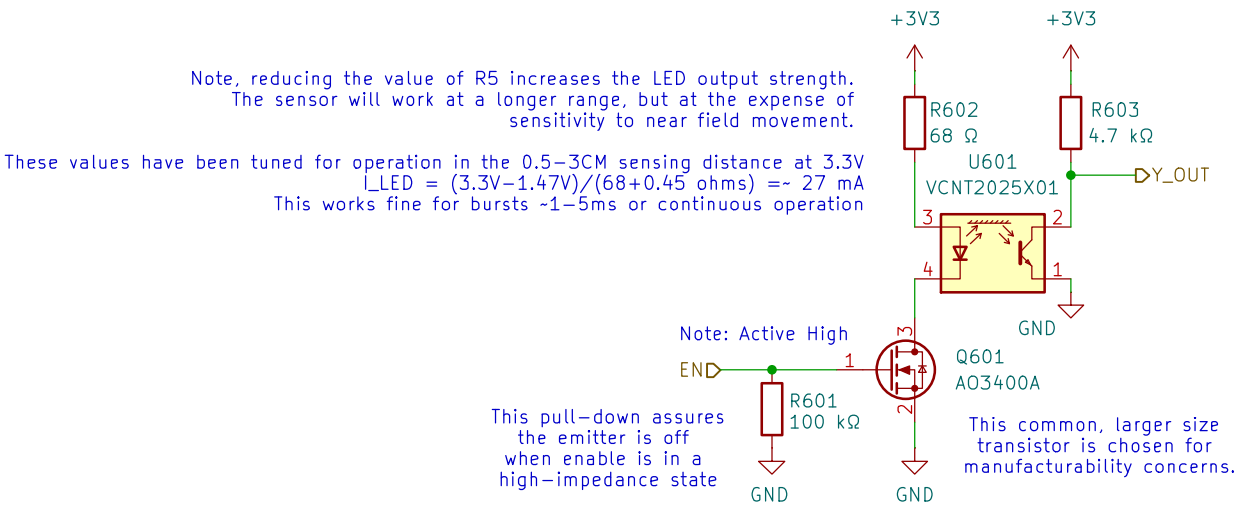
Date: 2026-01-02

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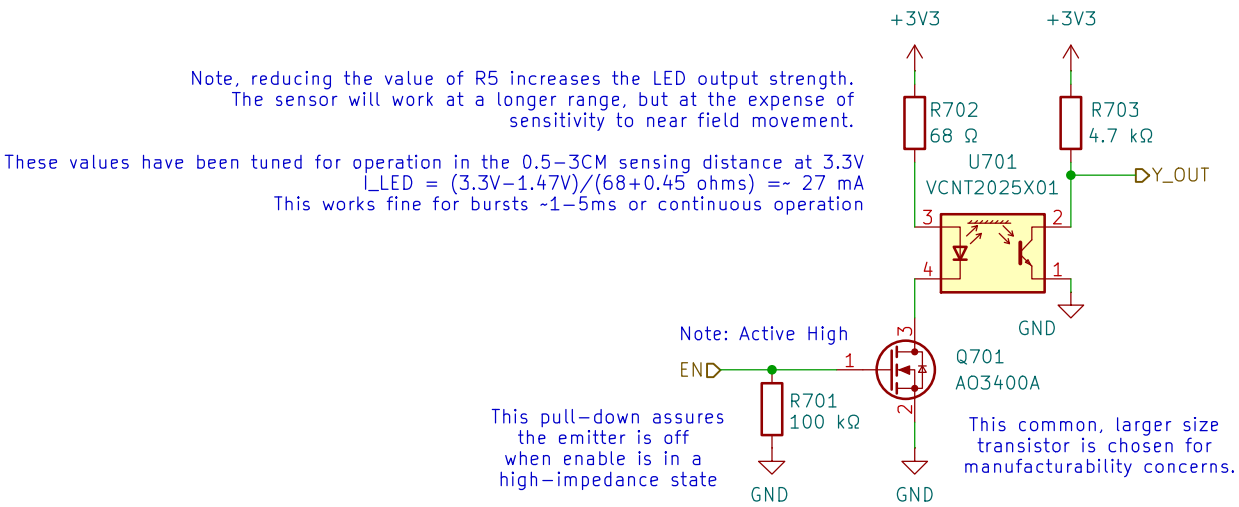


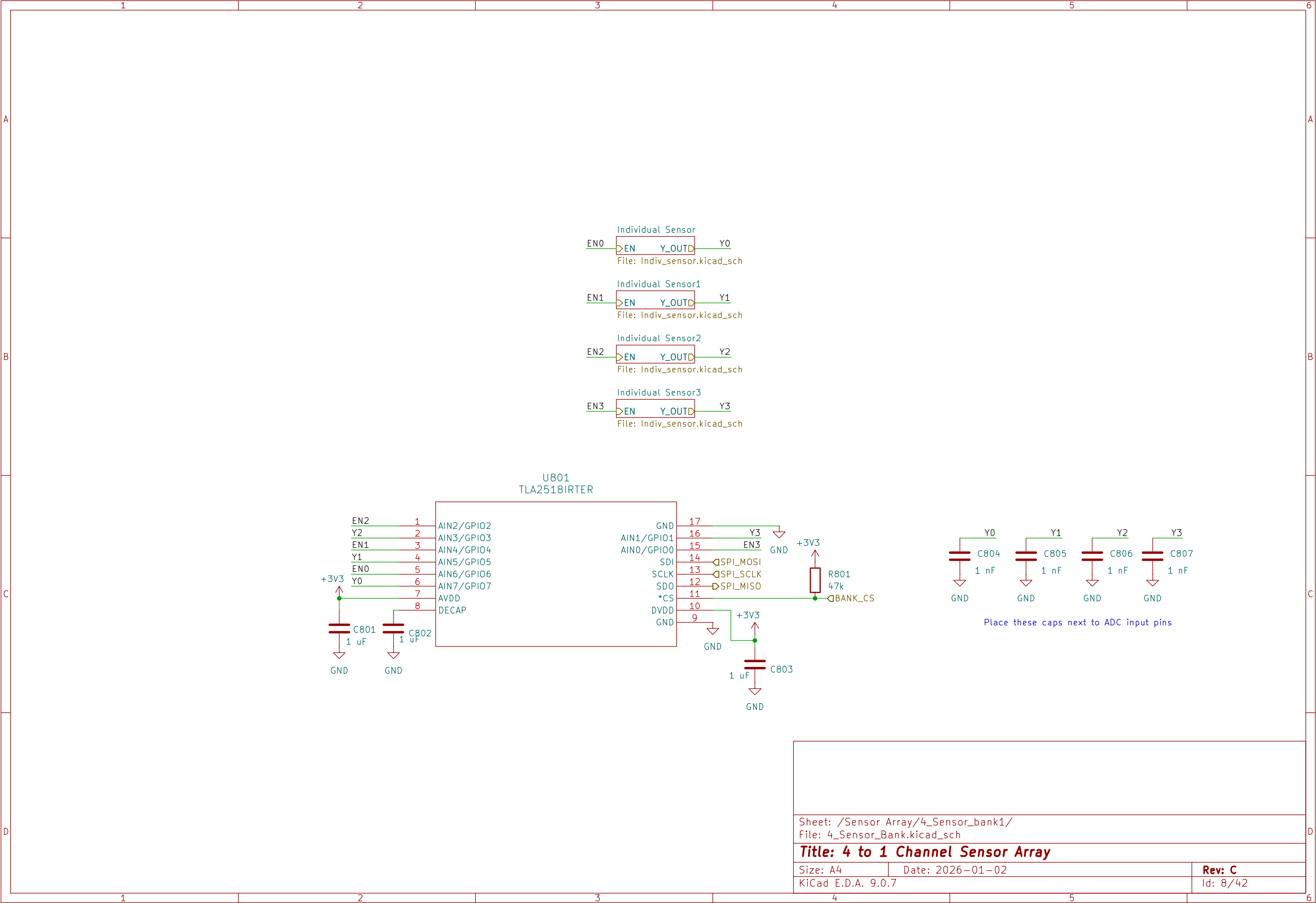


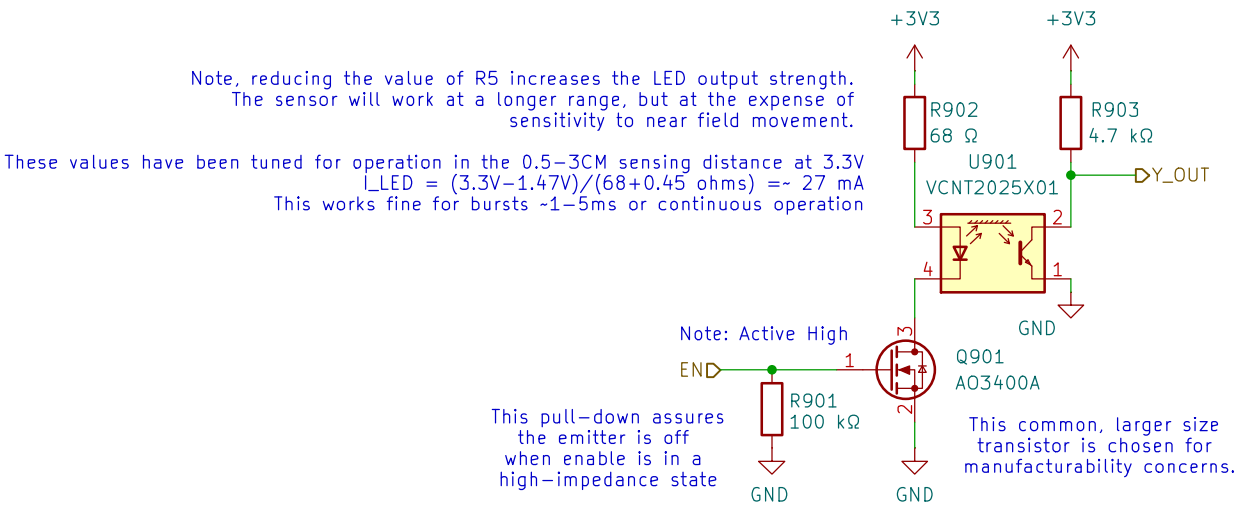


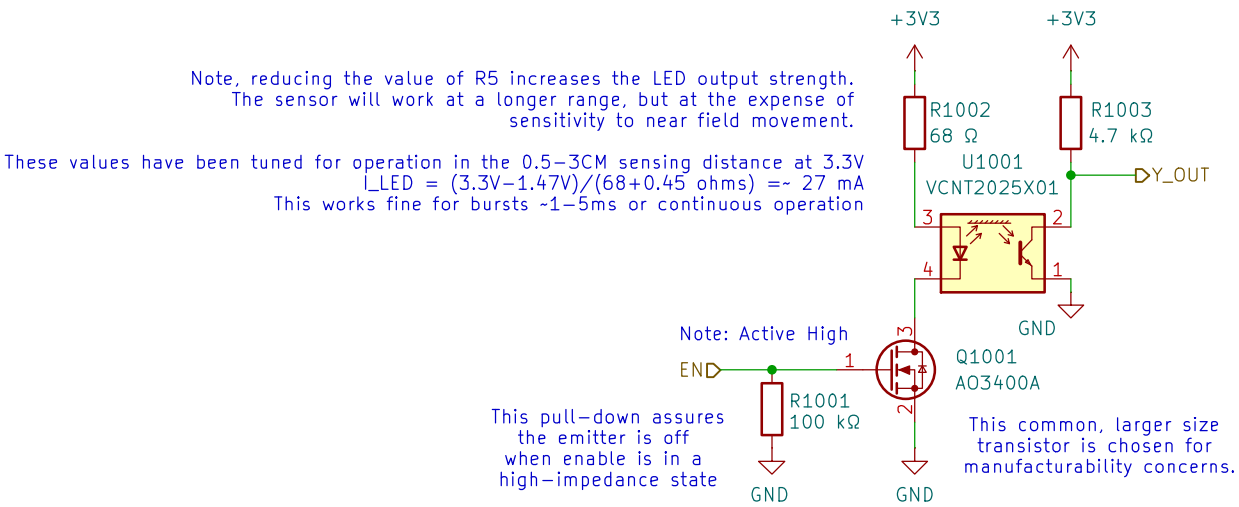


Sheet: /Sensor Array/4_Sensor_bank/Individual Sensor2/ File: Indiv_sensor.kicad_sch		
Title:		
Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 6/42

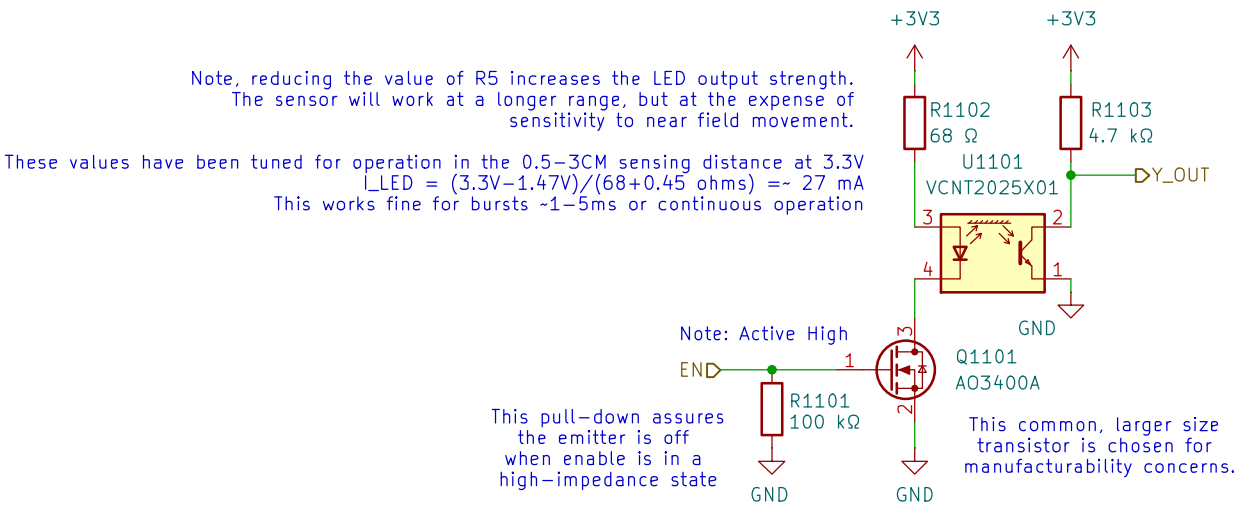




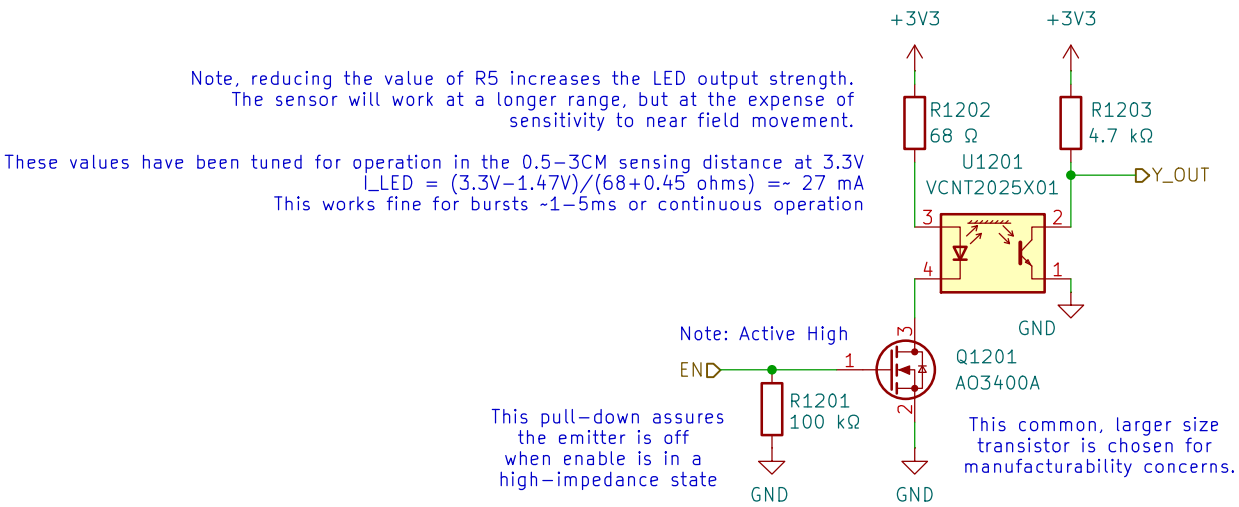




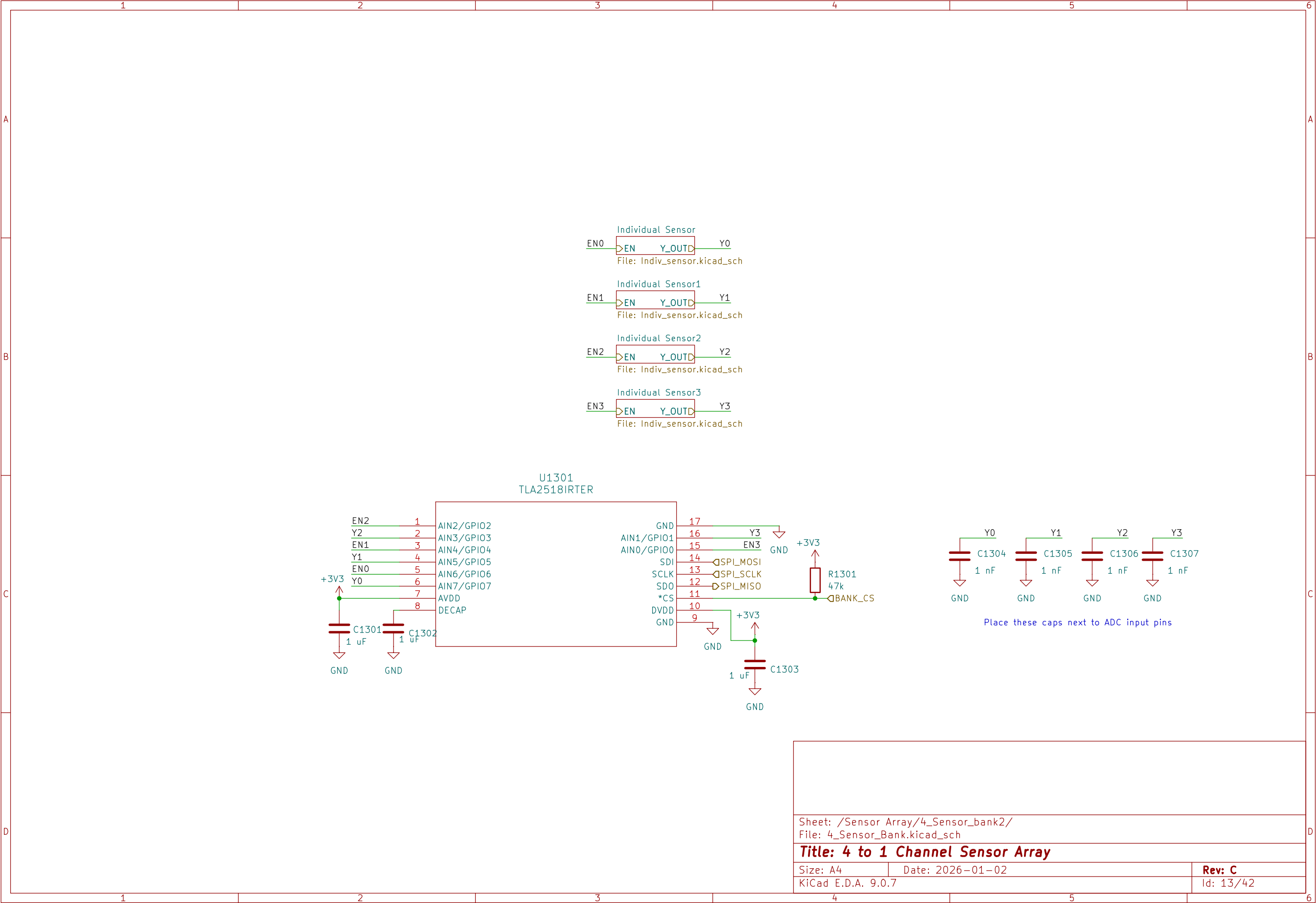
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KiCad E.D.A. 9.0.7		Id: 10/42

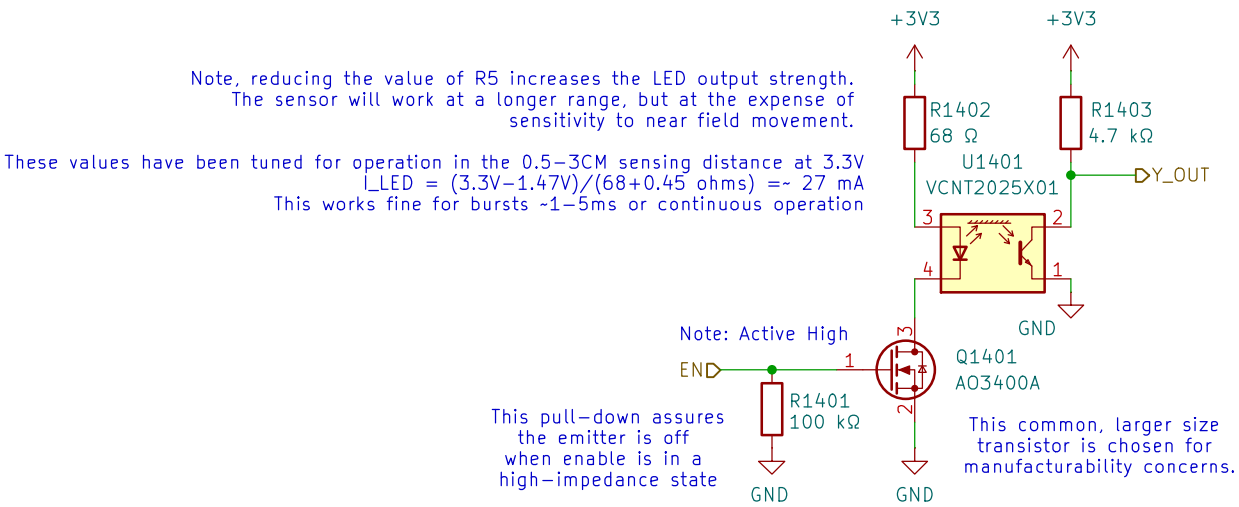


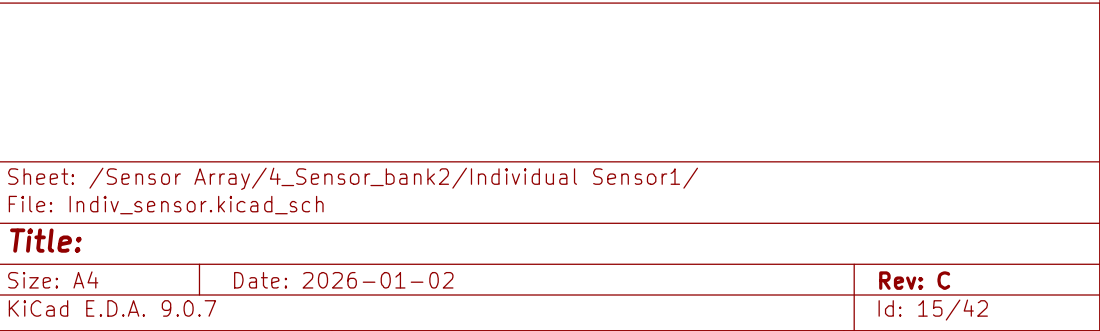
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KiCad E.D.A. 9.0.7		Id: 11/42

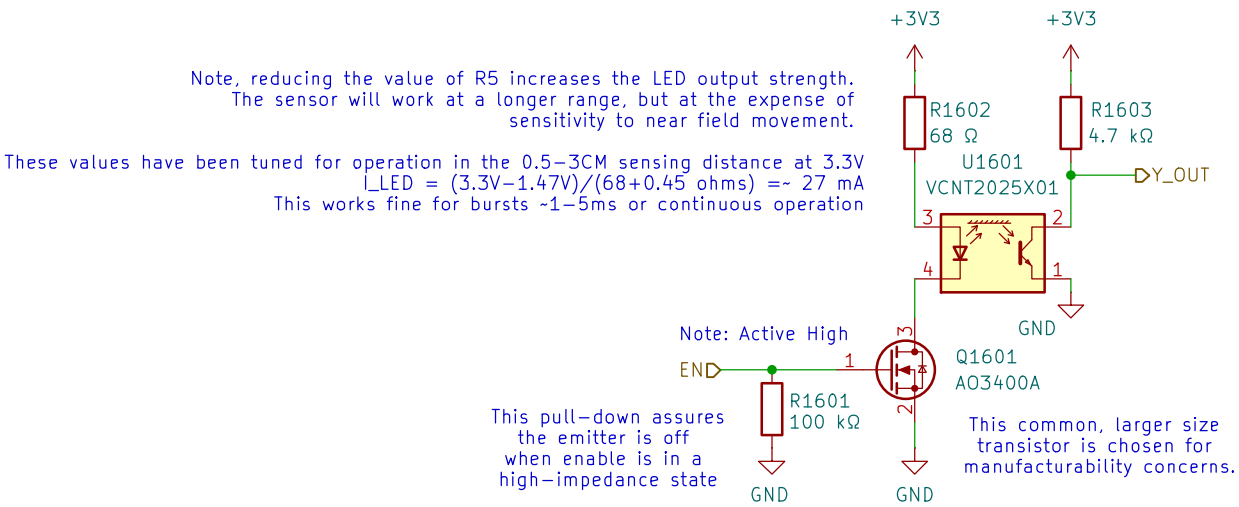


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Size: A4	Date: 2026-01-02	Rev: C
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Sheet: /Sensor Array/4_Sensor_bank2/Individual Sensor2/
File: Indiv_sensor.kicad_sch

Title:

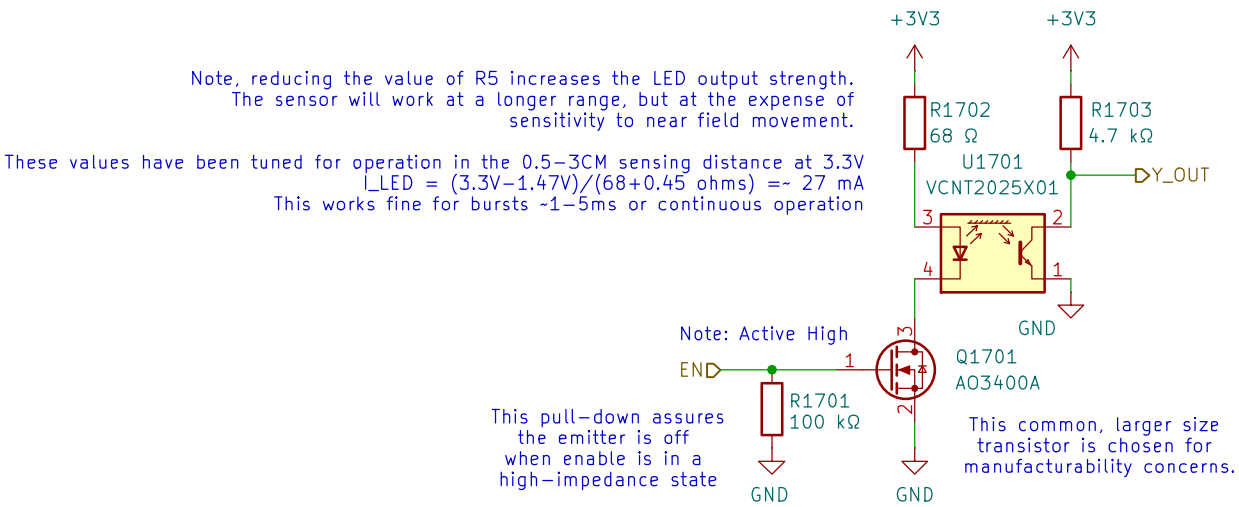
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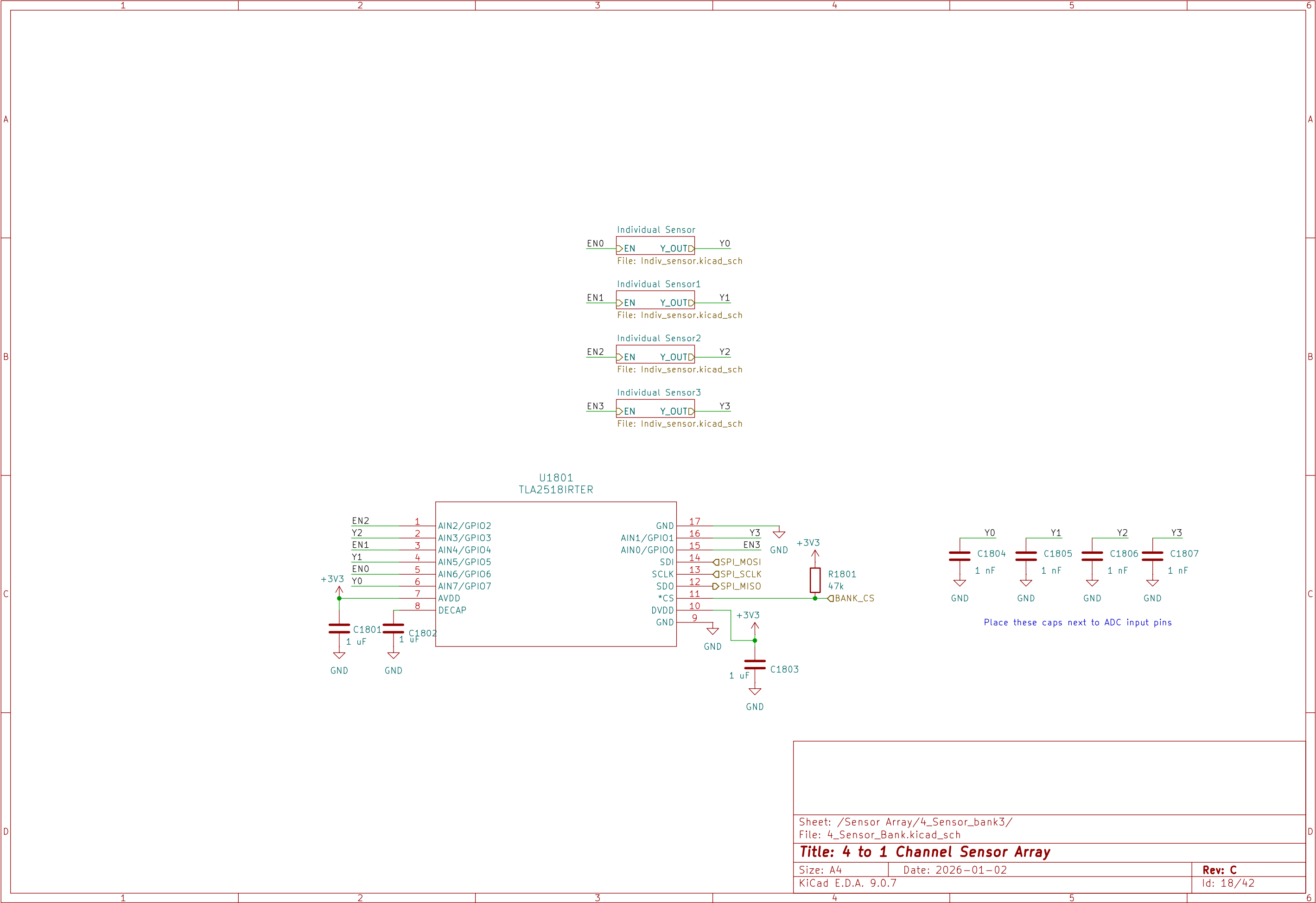
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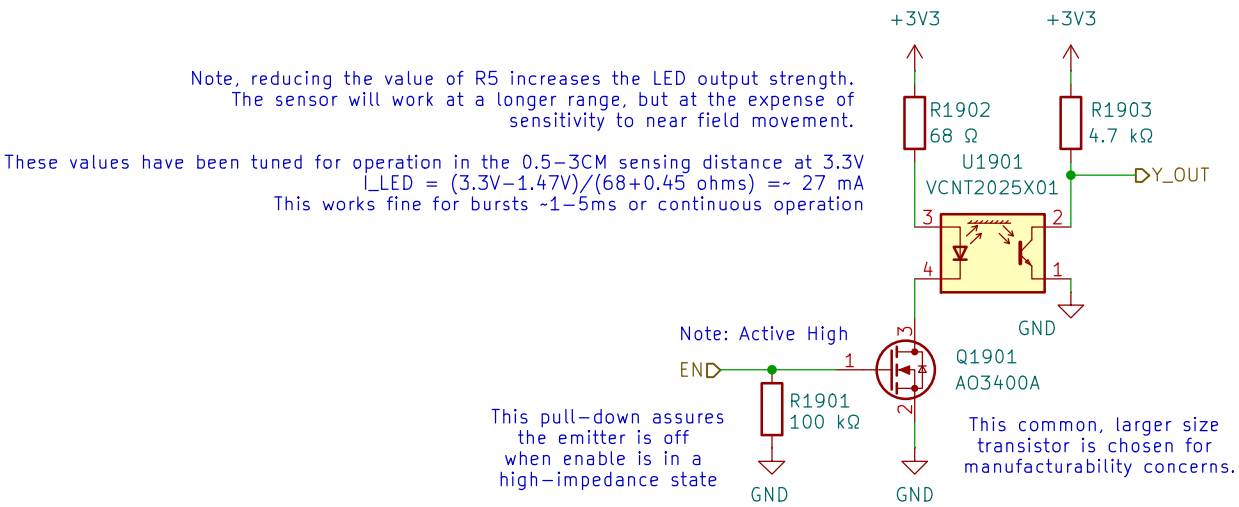
Rev: C

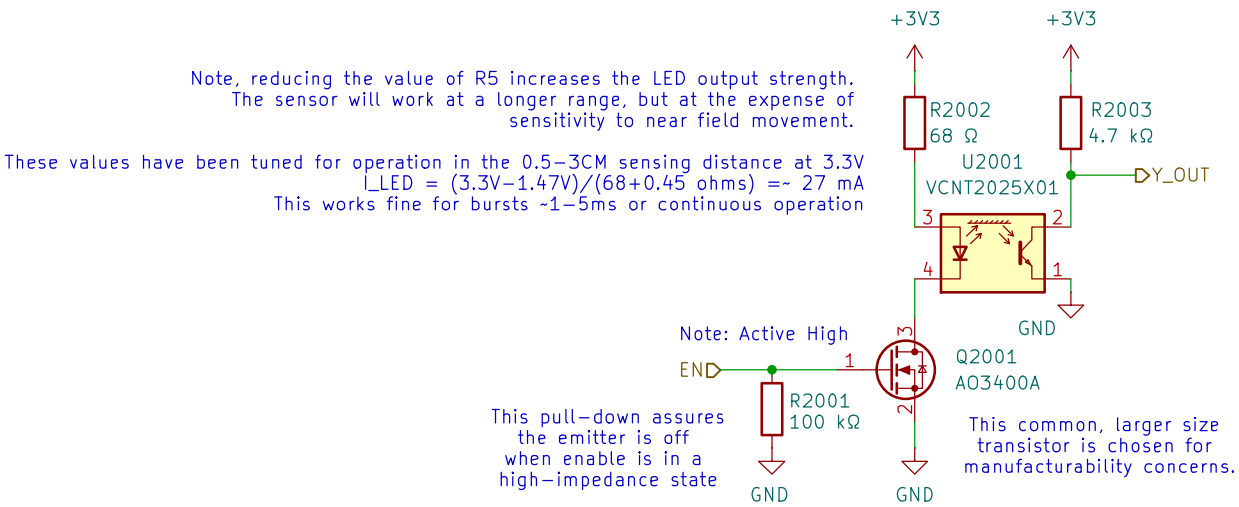
KiCad E.D.A. 9.0.7

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Sheet: /Sensor Array/4_Sensor_bank3/Individual Sensor1/
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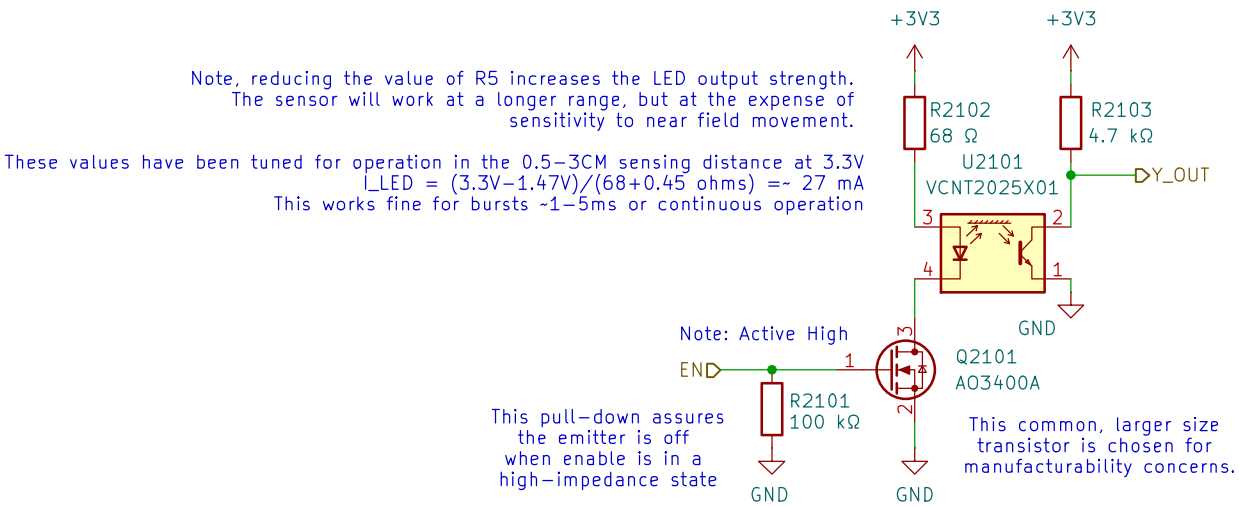
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Date: 2026-01-02

Rev: C

KiCad E.D.A. 9.0.7

Id: 20/42



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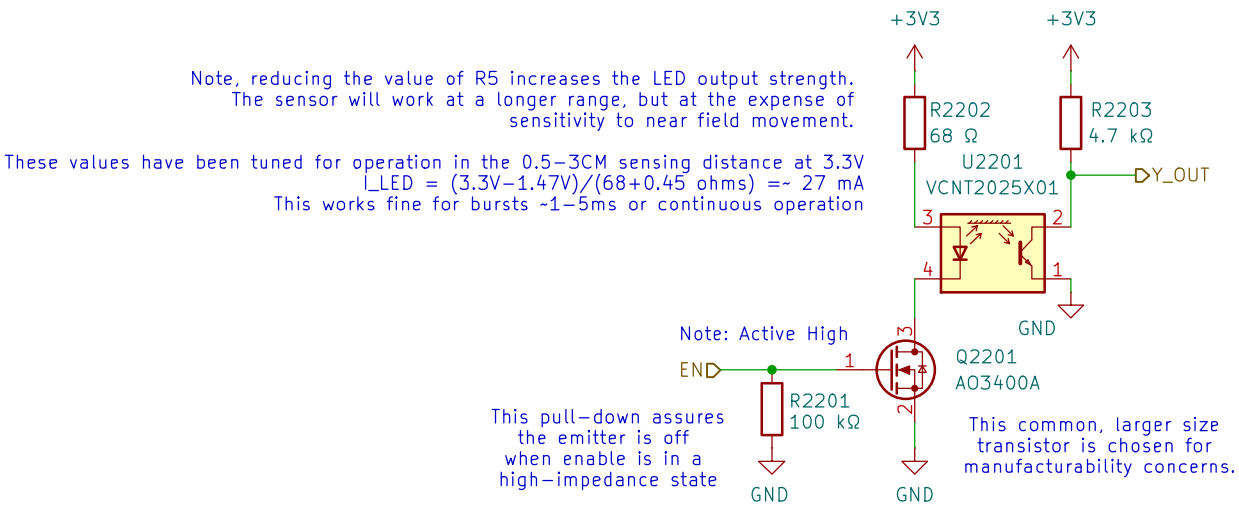
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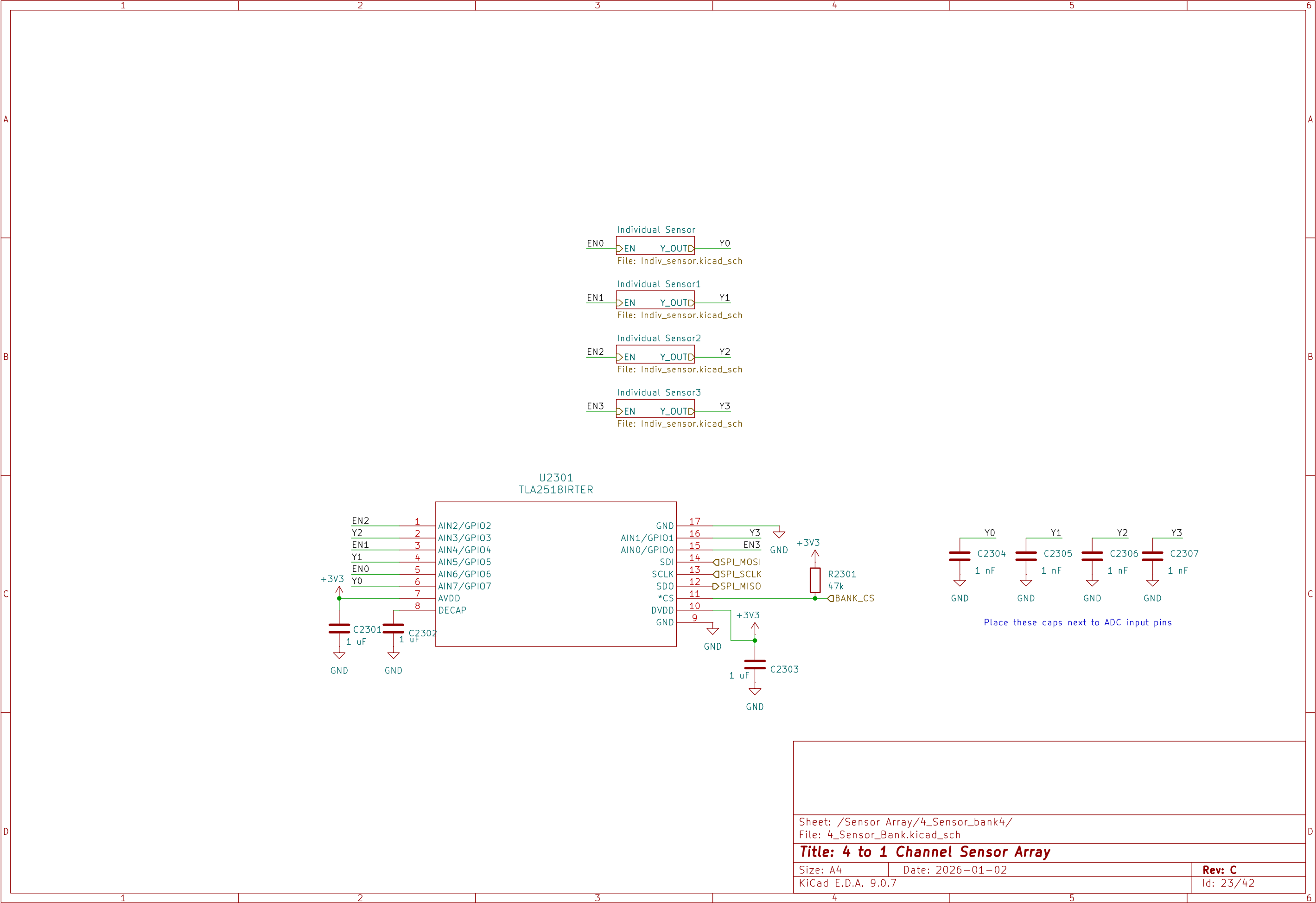
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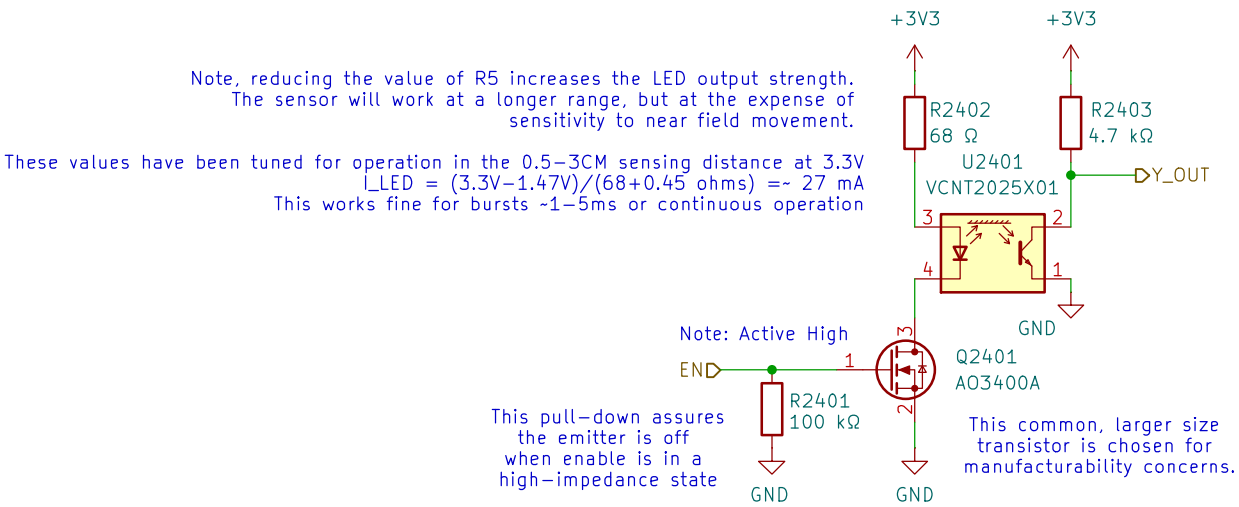
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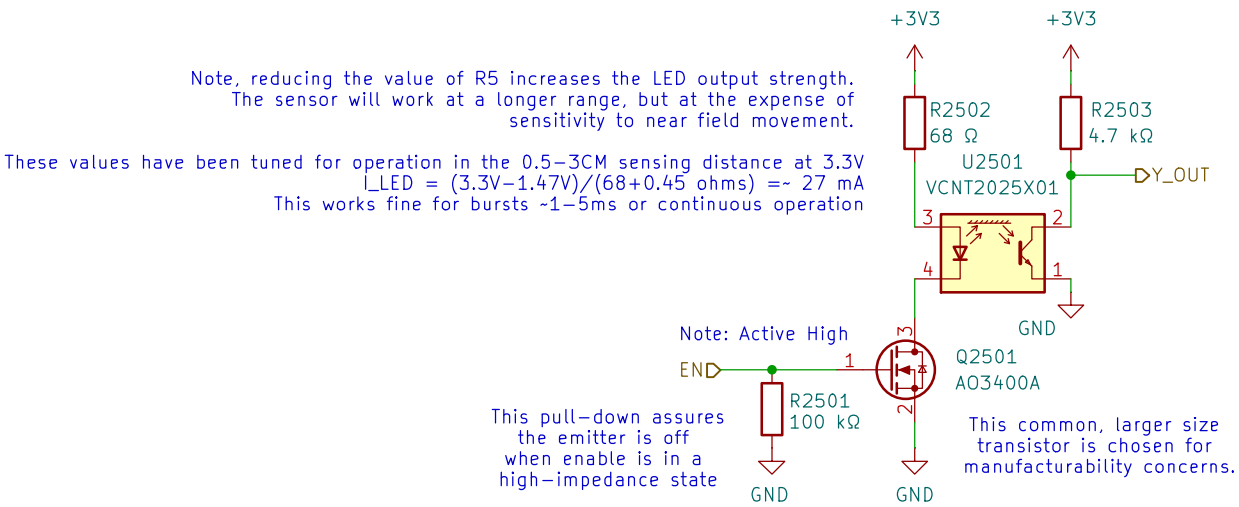
KiCad E.D.A. 9.0.7

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Sheet: /Sensor Array/4_Sensor_bank4/Individual Sensor1/
File: Indiv_sensor.kicad_sch

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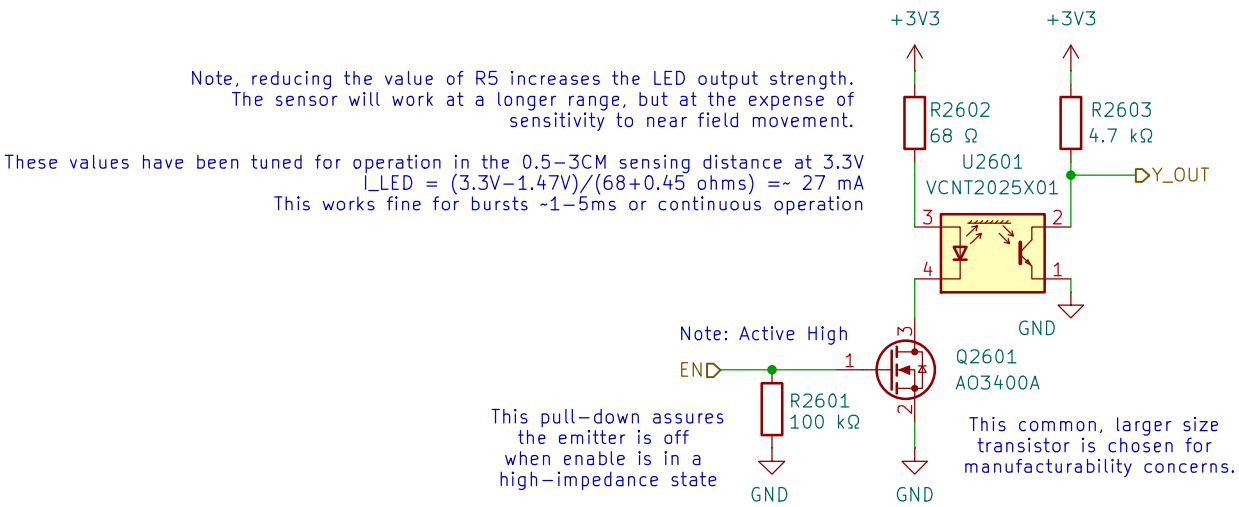
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Date: 2026-01-02

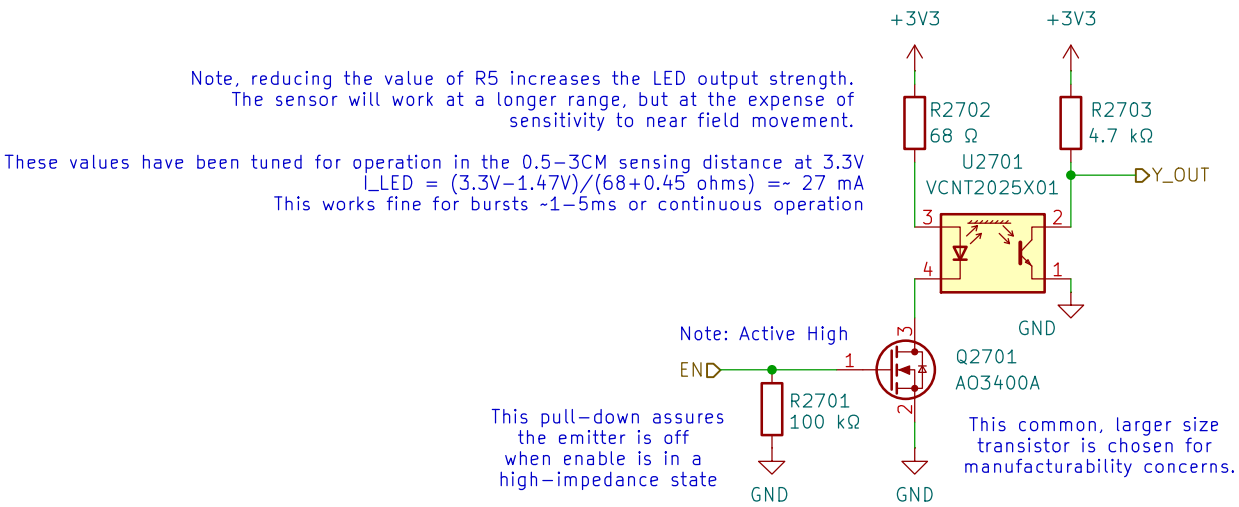
Rev: C

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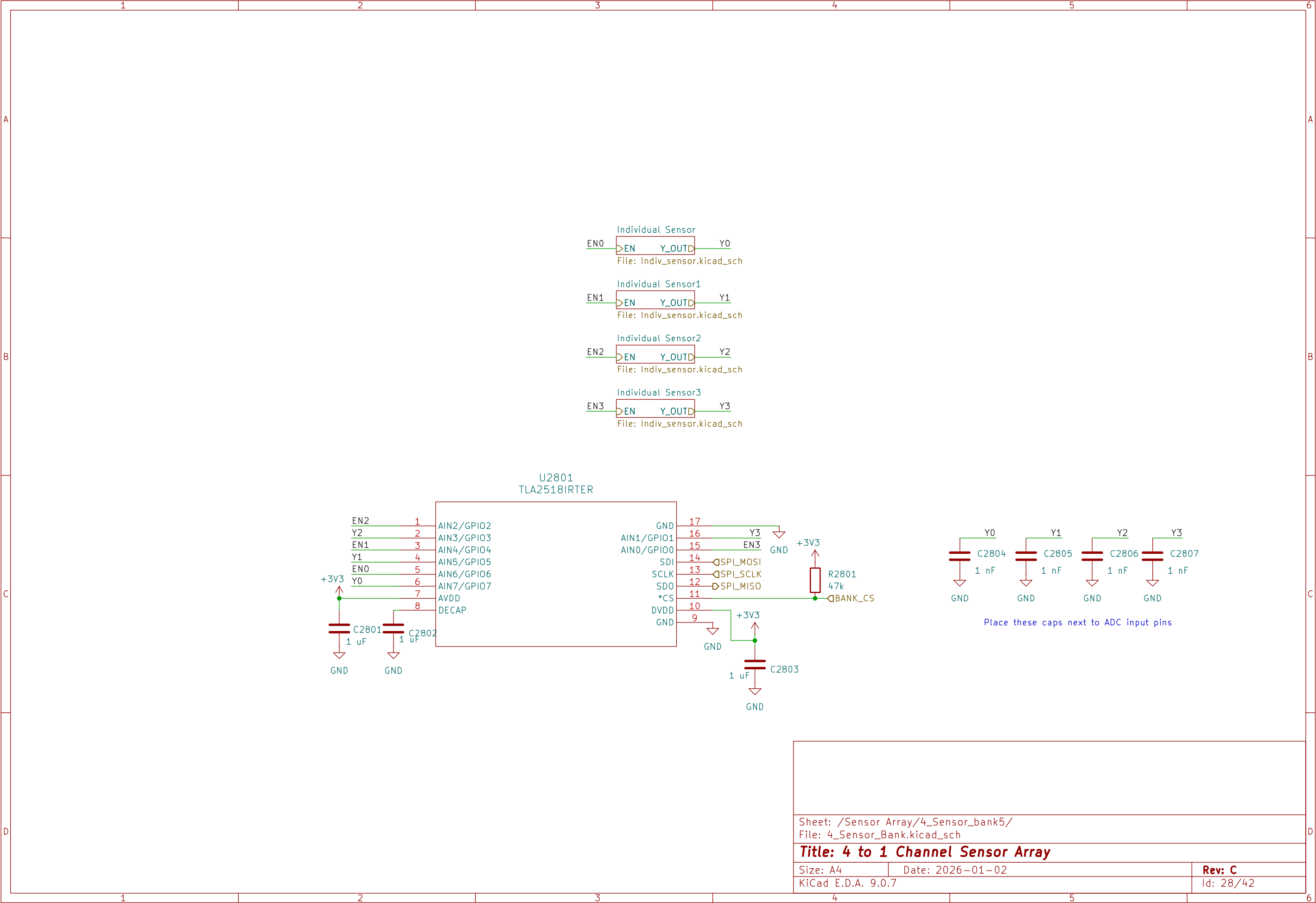
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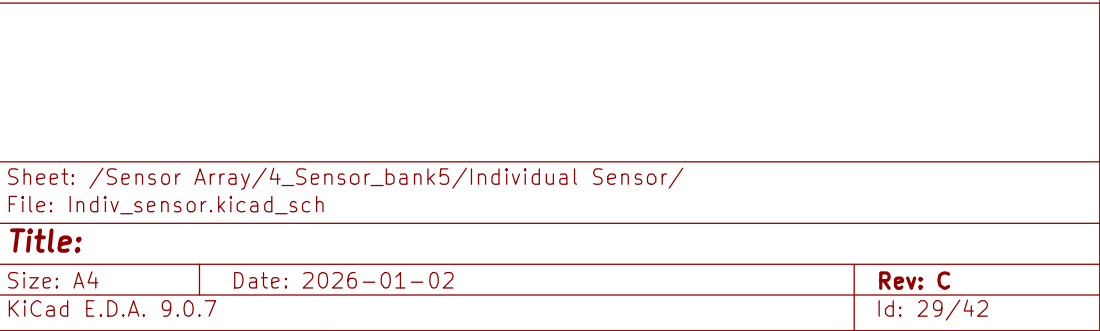


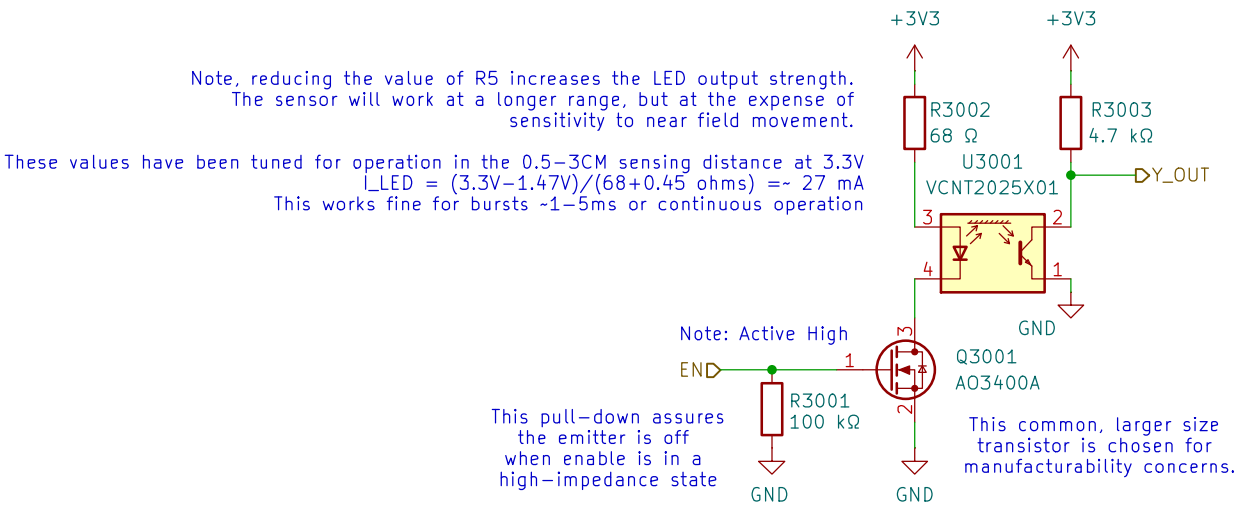
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Title:		
Size: A4	Date: 2026-01-02	Rev: C
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Sheet: /Sensor Array/4_Sensor_bank4/Individual Sensor3/		
File: Indiv_sensor.kicad_sch		
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Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 27/42







Sheet: /Sensor Array/4_Sensor_bank5/Individual Sensor1/
File: Indiv_sensor.kicad_sch

Title:

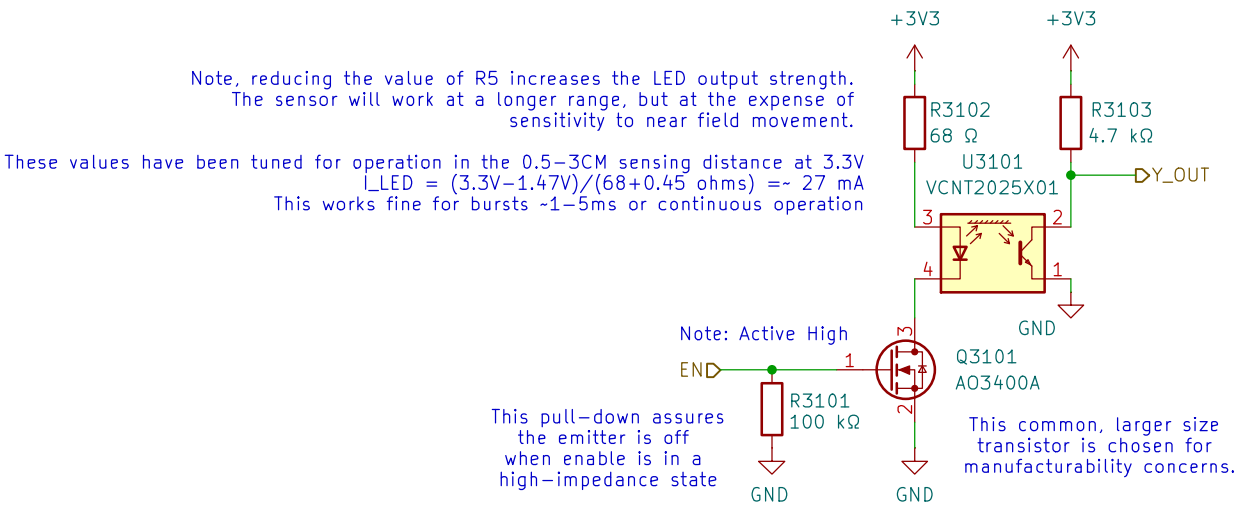
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Date: 2026-01-02

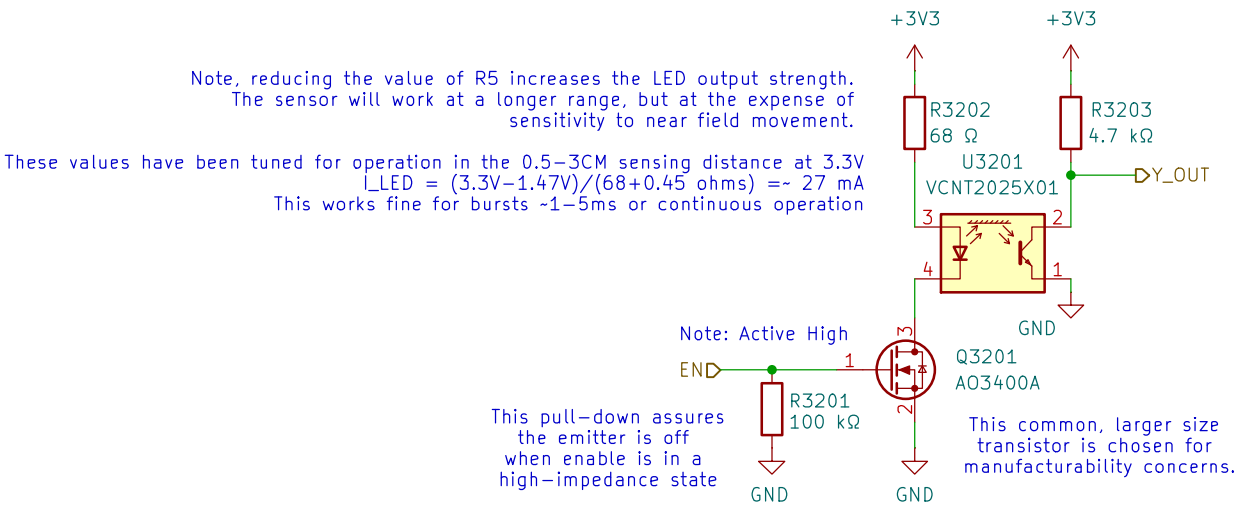
Rev: C

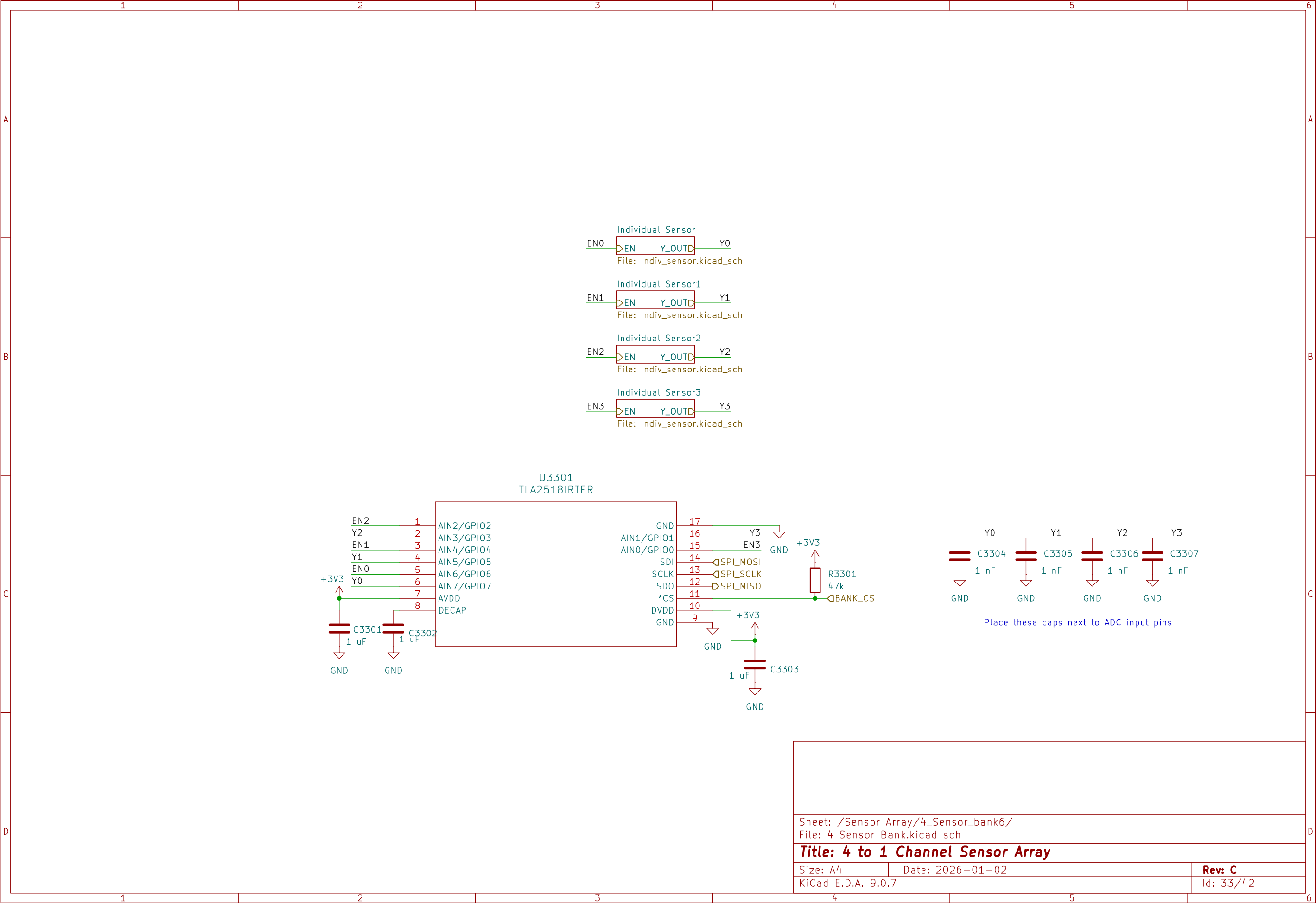
KiCad E.D.A. 9.0.7

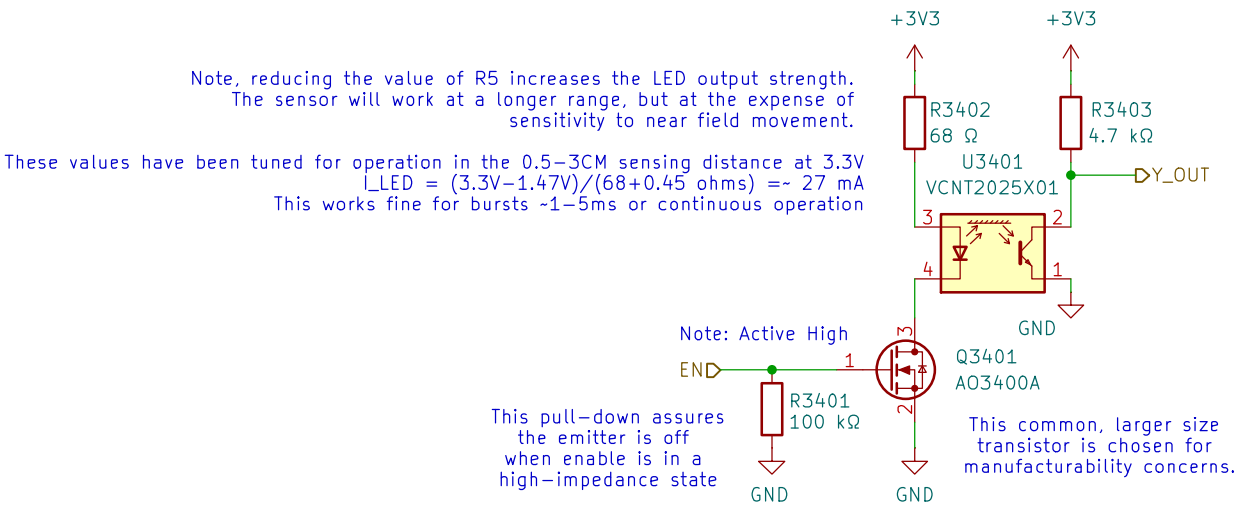
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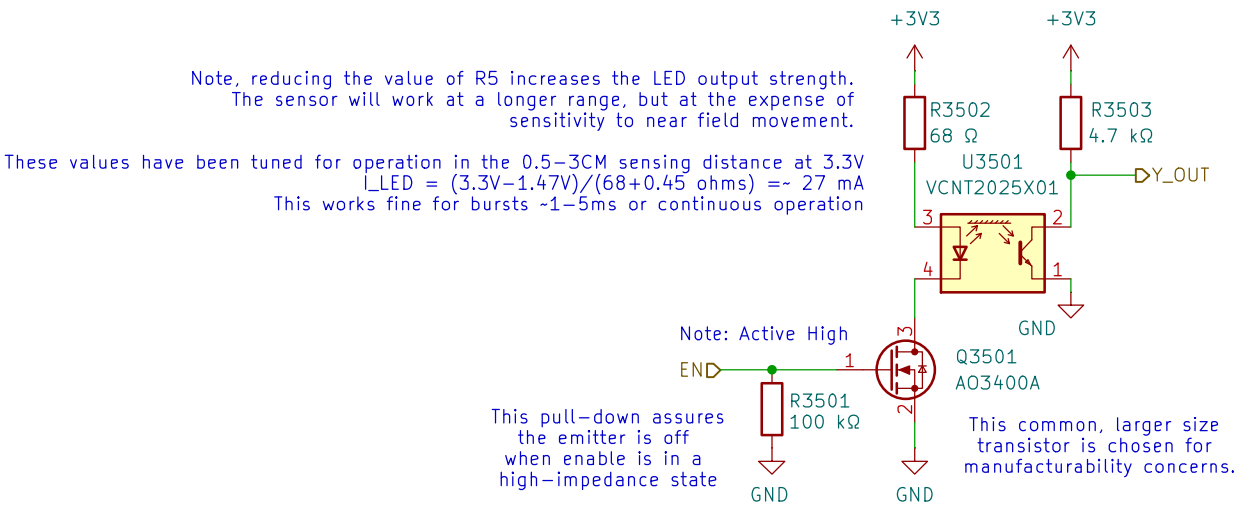


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Title:		
Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 31/42









Sheet: /Sensor Array/4_Sensor_bank6/Individual Sensor1/
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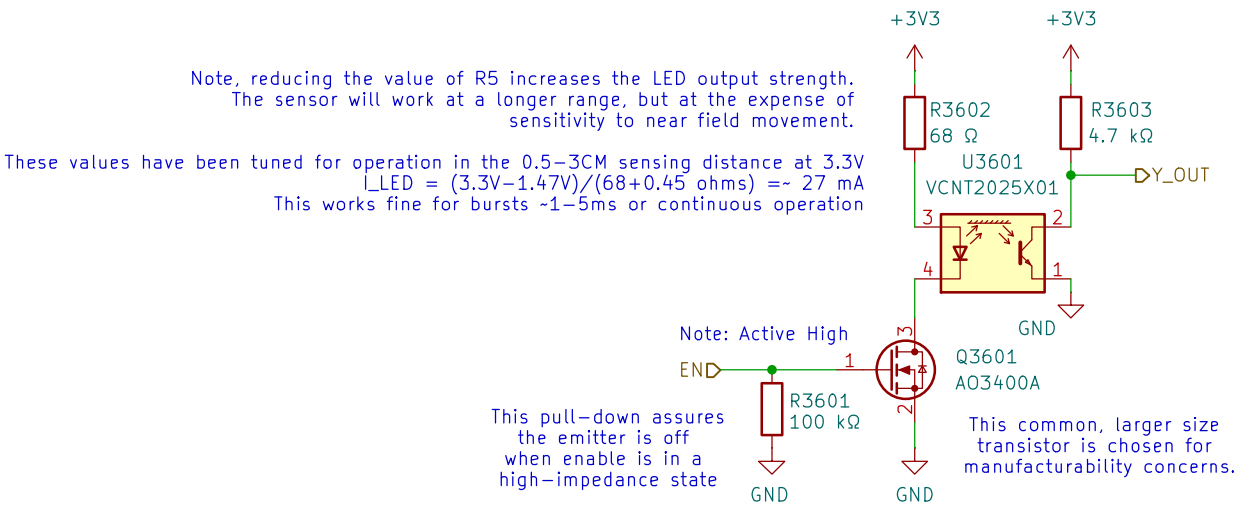
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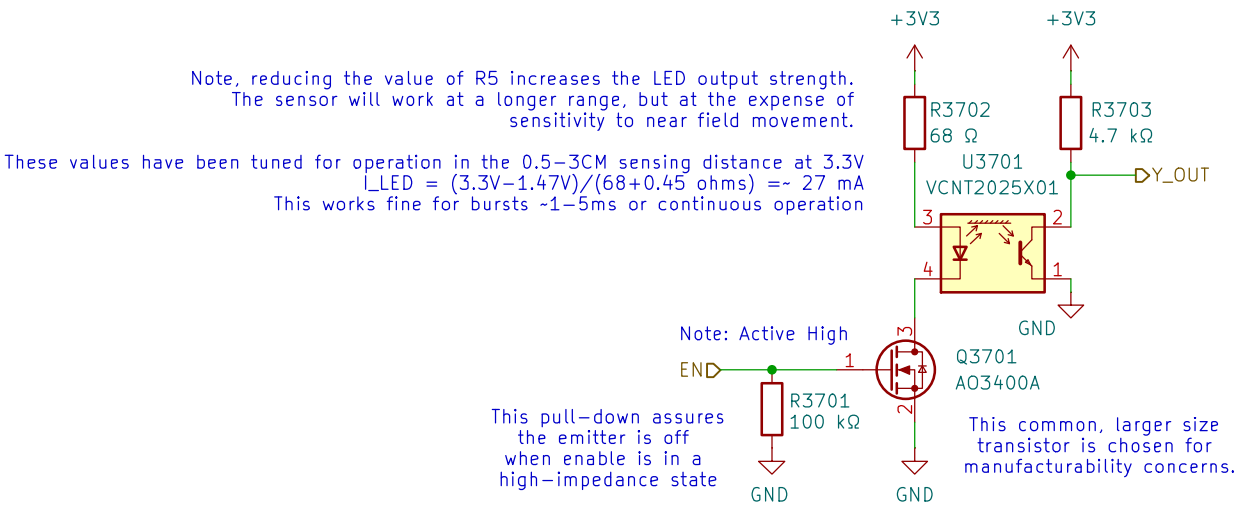
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Rev: C

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Sheet: /Sensor Array/4_Sensor_bank6/Individual Sensor3/
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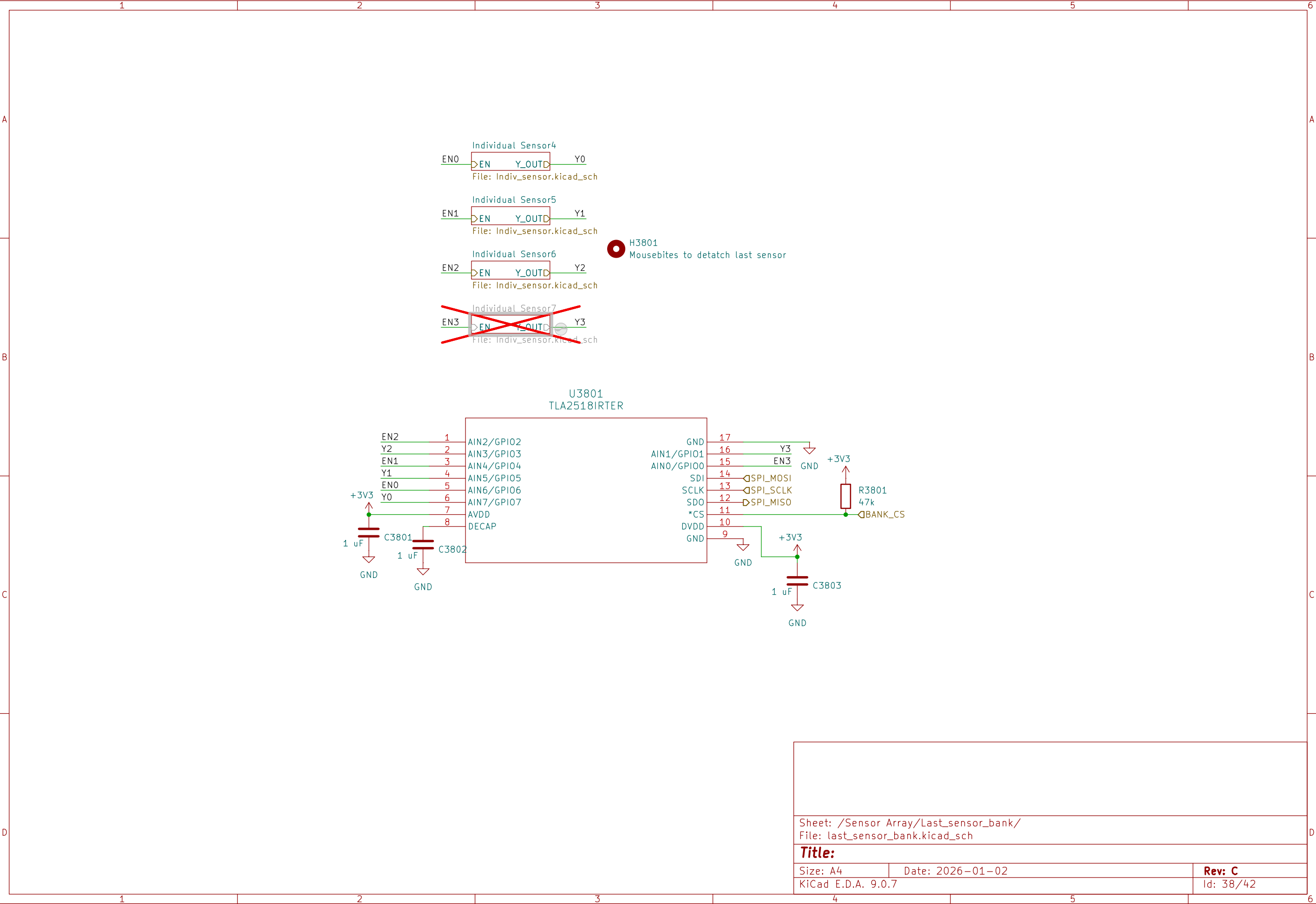
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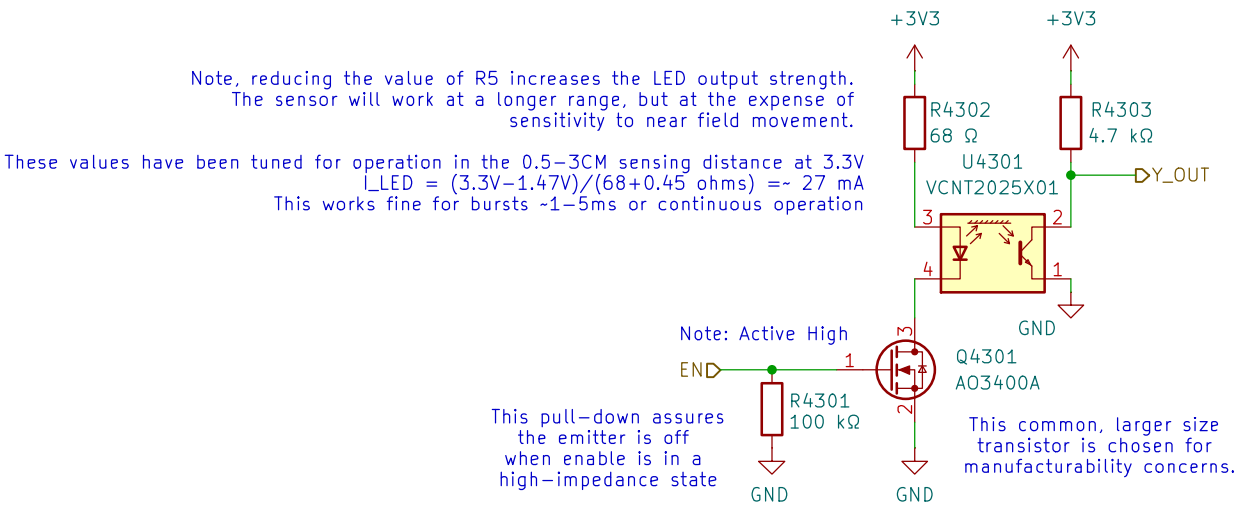
Date: 2026-01-02

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Sheet: /Sensor Array/Last_sensor_bank/Individual Sensor4/
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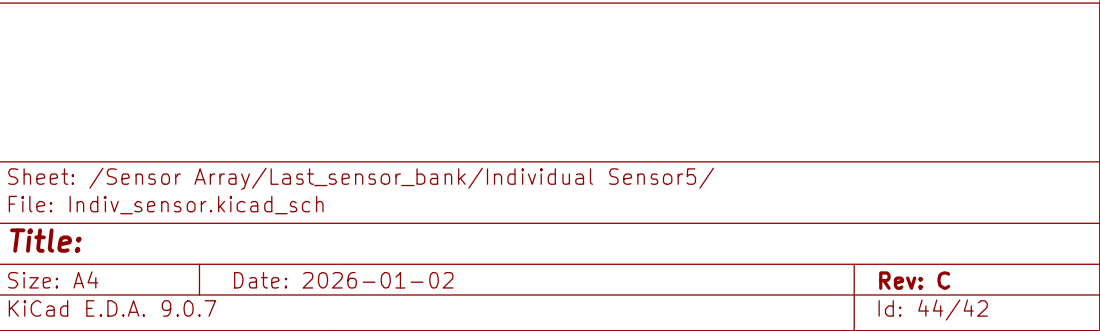
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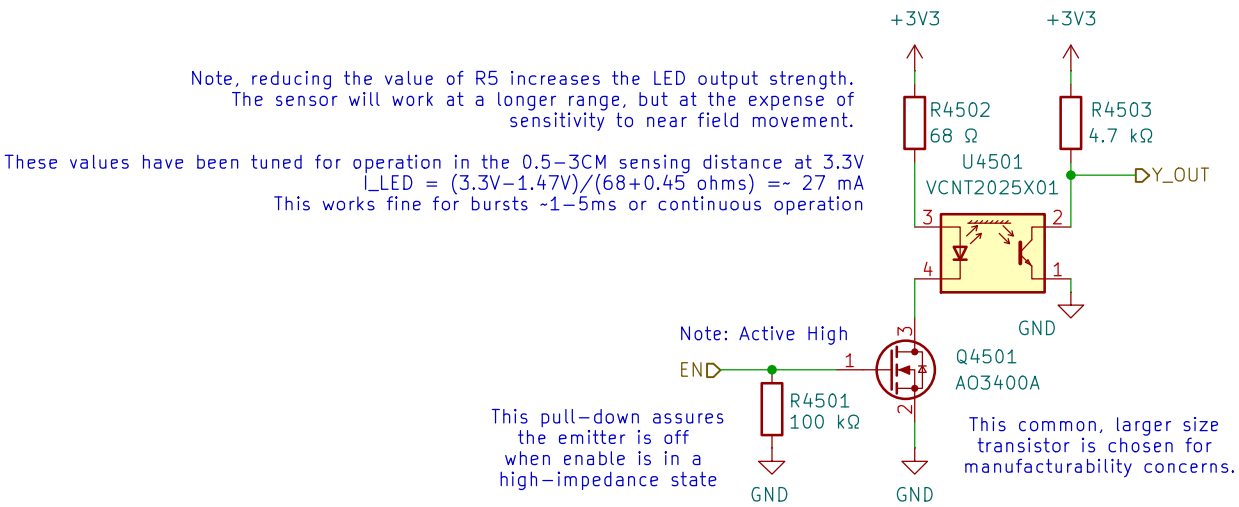
Date: 2026-01-02

Rev: C

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Sheet: /Sensor Array/Last_sensor_bank/Individual Sensor6/
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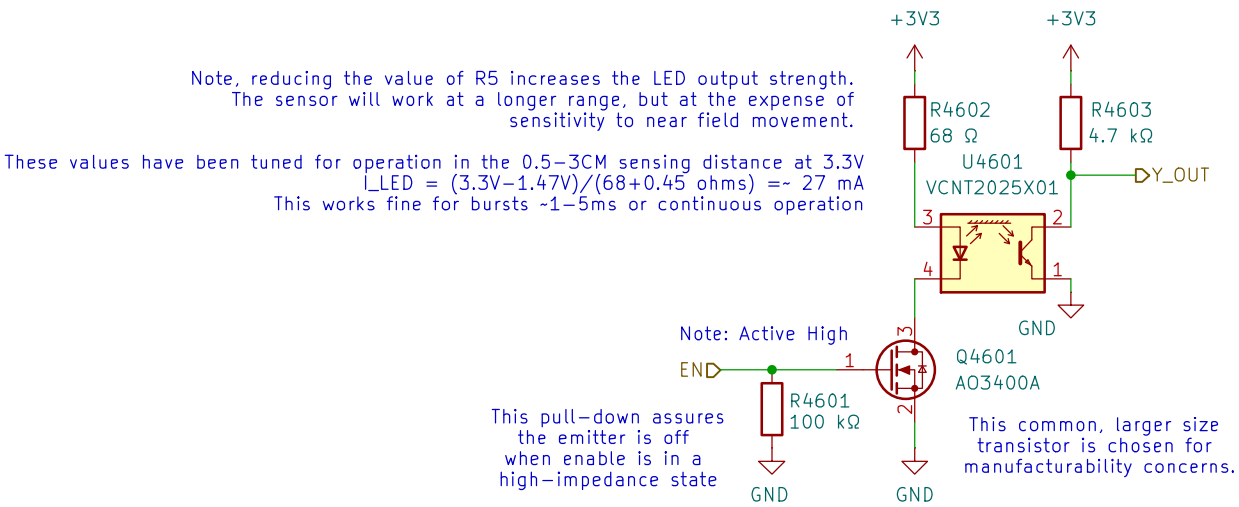
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Date: 2026-01-02

Rev: C

KiCad E.D.A. 9.0.7

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Sheet: /Sensor Array/Last_sensor_bank/Individual Sensor7/ File: Indiv_sensor.kicad_sch		
Title:		
Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 46/42